

SEO & Digital Analytics Dashboard

A Power BI solution designed for a private client to support SEO monitoring, digital marketing analysis and executive reporting.

Microsoft Power BI

Power Query

Star Schema

KPI Design

Digital Analytics

Project Overview

This project was developed for a private client to deliver a Business Intelligence solution for SEO and digital marketing analytics. Company information and business data have been anonymized to protect client confidentiality.

The objective was to transform raw digital marketing data into meaningful business insights through interactive dashboards, KPI reporting and dimensional data modeling in Microsoft Power BI.

The solution documents the complete BI workflow, from data preparation and modeling to dashboard design and executive reporting.

Business Challenge

The client required a centralized reporting solution to monitor website performance, search visibility, traffic quality and marketing effectiveness.

Website Performance

Monitor visits and platform usage over time.

SEO Visibility

Track search positions and keyword performance.

Traffic Quality

Analyze bounce rate and engagement trends.

Conversion

Measure how digital activity supports business outcomes.

Solution Architecture



Data Preparation

Power Query was used to prepare the data for analysis. The ETL process focused on data quality, consistency and reporting readiness.

- Imported and structured source data
- Standardized dates and numeric formats
- Filtered irrelevant and internal records
- Handled missing values
- Created calculated columns and business KPIs
- Validated the final dataset before reporting

Data Model

The analytical model follows a dimensional star schema. A central fact table stores measurable events and keys, while dimension tables support filtering and analysis.

Fact Table

Traffic, bounce rate, SEO position and conversion measures.

Date Dimension

Time-based analysis and monthly reporting.

Customer Dimension

Segment and regional analysis.

Website Dimension

Page and platform analysis.

SEO Dimension

Keyword and search visibility analysis.

Traffic Source Dimension

Channel and acquisition analysis.

Dashboard Features

- Monthly website traffic analysis
- Platform and device comparison
- Bounce rate monitoring
- SEO position trend analysis
- Traffic source comparison
- Conversion KPI tracking
- Geographic visualization
- Interactive filters for month, segment and keyword

Business Value

The dashboard enables stakeholders to:

- Monitor digital performance from one centralized view
- Identify changes in website traffic and SEO visibility
- Compare marketing channels and platform usage
- Evaluate visitor behaviour and conversion performance
- Reduce manual reporting effort
- Support faster, data-driven decisions

Skills Demonstrated

Business Intelligence

Executive reporting and decision support.

Power Query

Data cleaning, transformation and ETL.

Data Modeling

Dimensional modeling and star schema design.

KPI Development

Definition and calculation of business metrics.

Dashboard Design

Interactive visual reporting in Power BI.

Digital Analytics

SEO, traffic, bounce rate and conversion analysis.

Conclusion

This project demonstrates the complete Business Intelligence workflow, from data preparation and dimensional modeling to dashboard development and executive reporting.

The final solution transforms digital marketing data into actionable insights while maintaining client confidentiality and supporting scalable future development.